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Unit – 2 ⇝ Vector Space (Lecture Note)

Introduction

→ A vector space is a collection of objects called vectors together with two operations:

(1) Vector Addition

(2) Scalar Multiplication

These operations satisfy certain mathematical rules.

→ Vector space is one of the most important concept in linear algebra and is widely used in engineering, computer graphics, physics, machine learning, data science, communication system, artificial intelligence, navigation system, etc.

→ Vectors may be represented as numbers, ordered pairs, triplets, matrices, polynomials, or functions.

Symbol, Name and Elements of Different Sets

Symbol	Name and Elements of the Set		
$\mathbb{R}$	Set of all real numbers		
	x	y	z
$\mathbb{R}^+$	Set of all positive real numbers		
	x	y	z
$\mathbb{R}^2$	Set of all ordered pairs of real numbers		
	$u = (x_1, y_1)$	$v = (x_2, y_2)$	$w = (x_3, y_3)$
$\mathbb{R}^3$	Set of all ordered triples of real numbers		
	$u = (x_1, y_1, z_1)$	$v = (x_2, y_2, z_2)$	$w = (x_3, y_3, z_3)$
$\mathbb{R}^n$	Set of all ordered n-tuples of real numbers		
	$u = (x_1, x_2, x_3, \dots, x_n)$	$v = (y_1, y_2, y_3, \dots, y_n)$	$w = (z_1, z_2, z_3, \dots, z_n)$
$\mathbb{M}_2(\mathbb{R})$	Set of all 2×2 matrices over $\mathbb{R}$		
	$A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}$	$B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}$	$C = \begin{bmatrix} a_3 & b_3 \\ c_3 & d_3 \end{bmatrix}$
$\mathbb{M}_{2\times3}(\mathbb{R})$	Set of all 2×3 matrices over $\mathbb{R}$		
	$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ d_1 & e_1 & f_1 \end{bmatrix}$	$B = \begin{bmatrix} a_2 & b_2 & c_2 \\ d_2 & e_2 & f_2 \end{bmatrix}$	$C = \begin{bmatrix} a_3 & b_3 & c_3 \\ d_3 & e_3 & f_3 \end{bmatrix}$

$\mathbb{P}_2(\mathbb{R})$	Set of all real polynomials of degree at most 2		
	$p_1(x) = a_0 + a_1x + a_2x^2$	$p_2(x) = b_0 + b_1x + b_2x^2$	$p_3(x) = c_0 + c_1x + c_2x^2$
$\mathbb{P}_3(\mathbb{R})$	Set of all real polynomials of degree at most 3		
	$p_1(x) =$ $a_0 + a_1x + a_2x^2 + a_3x^3$	$p_2(x) =$ $b_0 + b_1x + b_2x^2 + b_3x^3$	$p_3(x) =$ $c_0 + c_1x + c_2x^2 + c_3x^3$

Method 1 ⇝ Vector Space

Real Vector Space

→ Let **V** be a non-empty set. Suppose two operations are defined on V:

1. Vector Addition (+)
2. Scalar Multiplication (·) over the **field** of real numbers $\mathbb{R}$.

If the following **ten axioms** are satisfied for arbitrary $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbf{V}$ & $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}$, then V is called a real vector space over the field $\mathbb{R}$ under the given operations.

Axioms

(1)	$u + v \in V$	(Closure under vector addition)
(2)	$u + v = v + u$	(Commutative law of addition)
(3)	$u + (v + w) = (u + v) + w$	(Associative law of addition)
(4)	$u + \mathbf{0} = u$	(Existence of zero vector in V)
(5)	$u + (-u) = \mathbf{0}$	(Existence of additive inverse in V)
(6)	$\alpha \cdot u \in V$	(Closure under scalar multiplication)
(7)	$\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$	(Distributive law over vector addition)
(8)	$(\alpha + \beta) \cdot u = \alpha \cdot u + \beta \cdot u$	(Distributive law over scalar addition)
(9)	$\alpha \cdot (\beta \cdot u) = (\alpha\beta) \cdot u$	(Associative law of scalar multiplication)
(10)	$1 \cdot u = u$	(Multiplicative identity)

If V is vector space under Vector Addition (+) and Scalar Multiplication (·), then it can be written in short as $(\mathbf{V}, +, \cdot)$ is a vector space.

Remarks:

(1) To prove a set is vector space, **all ten axioms** must be satisfied. If at least one axiom fails, the set is not a vector space.

(2) If the binary operations (+ or ·) are **not explicitly** defined, then for a vector space V, the operations are usually assumed to be the standard operations, namely ordinary **vector addition** and ordinary **scalar multiplication**.

Example:

Vector Addition : $(a, b) + (c, d) = (a + c,\ b + d)$

Scalar Multiplication : $k \cdot (a, b) = (ka,\ kb)$

(3) The first five axioms are related to **vector addition**, while the remaining five axioms are related to **scalar multiplication**.

(4) The **zero vector 0** in a vector space is **unique**.

(5) The **additive inverse** of each vector is also **unique**.

(6) A vector space may contain vectors of different forms such as: Numbers, Ordered pairs, Ordered triples, Matrices, Polynomials, and Functions.

Examples of Method-1: Vector Spaces

C	1	Show that the set $V = \mathbb{R}^+$ of all positive real numbers with operations $x + y = xy$ and $k \cdot x = x^k$ is a vector space.
C	2	Check whether the set $V = \{(x, y)/x, y \in \mathbb{R}\}$ is a vector space or not, with standard operations. i.e., $(a, b) + (c, d) = (a + c, b + d)$ and $k \cdot (a, b) = (ka, kb)$. **Answer: Yes, Vector Space**
C	3	Let V be the set of all ordered pairs of real numbers with vector addition defined as $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2 + 1, y_1 + y_2 + 1)$. Show that the first five axioms for vector addition are satisfied. Mention the zero vector and additive inverse. **Answer: Zero vector $= (-1, -1)$ &** **Additive Inverse $= (-x - 2, -y - 2)$**
C	4	The set V of all triples of real numbers (x, y, z) with the operations: $(x, y, z) + (x', y', z') = (x + x', y + y', z + z')$ & $k \cdot (x, y, z) = (kx, y, z)$. Determine whether the set V is a vector space or not under the given operations. **Answer: No, 8th condition is not satisfied**

C	5	Check the set of 2×2 matrices, $V = \left\{ \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} / a, b \in \mathbb{R} \right\}$ is a vector space or not under $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} + \begin{bmatrix} x & 1 \\ 1 & y \end{bmatrix} = \begin{bmatrix} a+x & 1 \\ 1 & b+y \end{bmatrix}$ and $k \cdot \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} = \begin{bmatrix} ka & 1 \\ 1 & kb \end{bmatrix}$. **Answer: Yes, Vector Space**
C	6	Check whether the set $\mathbb{P}_2 = \{a_0 + a_1x + a_2x^2 / a_0, a_1, a_2 \in \mathbb{R}\}$ is a vector space or not, with standard operations. **Answer: Yes, Vector Space**

Method 2 ⇝ Subspaces

Subspace of a Vector Space

→ A non-empty subset W of a vector space V (i.e., $\Phi \neq W \subseteq V$) is called a **subspace** of V if W is itself a vector space under the **same vector addition** and **scalar multiplication** defined on V.

→ In other words, let V be a vector space with the binary operations '+' and '·'. Let W be a non-empty subset of V. Then W is said to be a subspace of V if the following conditions are satisfied for all $u, v \in W$ and $k \in \mathbb{R}$:

(1) $u + v \in W$ (Closure under vector addition)

(2) $k \cdot u \in W$ (Closure under scalar multiplication)

Remarks

(1) Every subspace of a vector space is itself a vector space under the same operations inherited from the parent vector space.

(2) Every subspace **must** contain the **zero vector** of V.

(3) The trivial subspaces of a vector space V are { **0** } and **V** itself.

(4) A non-empty subset W of V is a subspace of V if and only if $\forall\, u, v \in W$ and $k \in \mathbb{R}$, $\mathbf{k \cdot u + v \in W}$.

Examples of Method-2: Subspaces

C	1	Prove that $W = \{(x, y) \in \mathbb{R}^2 \,/\, y = mx,\ m \in \mathbb{R}\}$ is subspace of $\mathbb{R}^2$.
C	2	Is $W = \{(x, y) \,/\, x^2 = y^2\}$ a subspace of $\mathbb{R}^2$? **Answer: No, 1st condition is not satisfied**
C	3	Check whether $W = \{(x, y, z) \,/\, ax - by + cz = 0\}$ is a subspace of $\mathbb{R}^3$ or not? **Answer: Yes, Subspace**
C	4	Prove that $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \,/\, 2a + 3b - c + d = 0 \right\}$ is a subspace of $\mathbb{M}_2$.
C	5	Is $W = \{a_0 + a_1x + a_2x^2 + a_3x^3 \,/\, a_0, a_1, a_2, a_3 \in \mathbb{Z} \text{ and } x \in \mathbb{R}\}$ is a subspace of $\mathbb{P}_3$? **Answer: No, 2nd conditions are not satisfied**

Method 3 ⇝ Linear Combination and Span

Linear Combination

→ Let $S = \{u_1, u_2, u_3, \dots, u_n\} \subseteq V$, and let $c_1, c_2, c_3, \dots, c_n \in \mathbb{R}$ be scalars. The expression of the form $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n}$ is called a **linear combination** of the vectors of S.

→ A vector v can be written as linear combination of the vectors $u_1, u_2, u_3, \dots, u_n$ if there exist scalars $c_1, c_2, c_3, \dots, c_n \in \mathbb{R}$ such that $\mathbf{v = c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n}$.

Procedure to a check v can be written as linear combination of $u_1, u_2, \dots, u_n$

(1) Express v as a linear combination of the given vectors:

i.e. $c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = v$.

(2) Form the corresponding system of linear equations $AX = B$.

$$\text{Where } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m\times n} \quad ; u_i = (a_{i1}, a_{i2}, a_{i3}, \dots, a_{in})$$

$$X = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{bmatrix}$$

$$B = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix} \quad ; \quad v = (v_1, v_2, v_3, \dots, v_n)$$

(3) Solve the system and analyze the consistency of the system:

- If the system is consistent **(i.e., at least one solution exists)**, then v is a linear combination of $u_1, u_2, u_3, \dots, u_n$.
- If the system is inconsistent **(i.e., no solution exists)**, then v is not a linear combination of $u_1, u_2, u_3, \dots, u_n$.

Remarks

→ To analyze the consistency of the system $AX = B$ **(For Square Matrix only)**

- If $\mathbf{det\,(A) \neq 0}$, then the system has a unique solution. Hence, **v is a linear combination of $\mathbf{u_1, u_2, u_3, \dots, u_n}$.**
- If $\mathbf{det(A) = 0}$, then the system have infinitely many solutions or no solution. So, **there is no conclusion**.

Span of a Set of Vectors

→ Let V be a vector space and $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a finite subset of V. The span of S, denoted by Span(S), is defined as the set of all possible linear combinations of the vectors in S.

$$\text{i.e.,}\quad \text{Span}(S) = \{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n / c_1, c_2, c_3, \dots, c_n \in \mathbb{R}\}.$$

→ The span of S consists of all vectors that can be generated using the vectors of S.

Spanning Set

→ Let V be a vector space and $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a finite subset of V. The set S is called a spanning set of V if every vector in V can be expressed as a linear combination of the vectors in S.

i.e. $V = \text{Span}(S) = \{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n / c_1, c_2, c_3, \dots, c_n \in \mathbb{R}\}$.

Procedure to check whether S spans the vector space V

→ Let $S = \{u_1, u_2, u_3, \dots, u_n\}$.

(1) Choose an arbitrary vector $b \in V$.

(2) Express b as a linear combination of the vectors of S:

i.e., $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = b}$.....(1)

(3) Form the corresponding system of linear equations and check its consistency.

- If the system of equations in (1) is consistent for all choices of $b \in V$,

  then $S = \{u_1, u_2, u_3, \dots, u_n\}$ spans the vector space V.

  i.e., $\mathbf{Span(S) = V.}$
- If the system of equations in (1) is inconsistent for some choice of $b \in V$,

  then $S = \{u_1, u_2, u_3, \dots, u_n\}$ does not span V.

Remarks

(1) The **span of a set** of vectors is always a **subspace** of the vector space V.

(2) The **span** of S is the **smallest subspace** of V containing all the vectors of S.

(3) If Span(S) = V, then the set S is called a **spanning set** of V.

(4) Different sets of vectors may have the same spanning set.

(5) If a vector $v \in S$ can be written as a linear combination of the remaining vectors of S, then removing v from S does not change the span.

Examples of Method-3: Linear Combination and Span

C	1	Express the vector $u = (-1, 1, 10)$ as a linear combination of the vectors $a = (1, 0, 1), b = (-2, 3, -2)$ & $c = (-6, 7, 5)$. **Answer: Yes, u is a LC of a, b, and c. [$u = 1a + (-2)b + 1c$]**
C	2	Express $u = (-5, 11, -7)$ as a linear combination of the vectors $a = (1, -2, 2)$, $b = (0, 5, 5)$ and $c = (2, 0, 8)$, if possible. **Answer: Not linear combination**
C	3	Check matrix A is a linear combination of A_1, A_2 and A_3 or not, where $A = \begin{bmatrix} 6 & -8 \\ -1 & -8 \end{bmatrix}$, $A_1 = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $A_3 = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$. **Answer: $A = A_1 + 2A_2 - 3A_3$**
C	4	Determine whether $v_1 = (2, 2, 2), v_2 = (0, 0, 3)$ and $v_3 = (0, 1, 1)$ span the vector space $\mathbb{R}^3$. **Answer: Yes**
C	5	Does $S = \{(1, -2, 2), (0, 5, 5), (2, 0, 8)\}$ span $\mathbb{R}^3$? Justify with reason. **Answer: No**
C	6	Check whether the following polynomials span $\mathbb{P}_2$ or not. $p_1(x) = 2 + x^2$, $p_2(x) = 1 - x + 2x^2$, $p_3(x) = 2 + x$ and $p_4(x) = 4 + x + x^2$. **Answer: Yes**

Method 4 ⇝ Linear Dependence and Independence

Linearly Dependent Set and Linearly Independent Set

→ Let $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a set of vectors from V.

Consider the equation $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = 0}$, where $c_1, c_2, \dots, c_n \in \mathbb{R}$.

- If above system has only the **trivial (unique) solution**, then S is said to be a **linearly independent (LI) set**.
- If above system has **nontrivial solution (infinite solution)**, then S is said to be **linearly dependent (LD) set**.

Remarks

(1) If a set of vectors $S = \{u_1, u_2, u_3, \dots, u_n\}$ from V contains the **zero vector**, then S is **linearly dependent**.

Example: $S = \{(1, 2, -1), (0, 0, 0), (-1, 2, -1)\}$ is linearly dependent.

(2) A **singleton set** containing a **nonzero vector** is linearly independent.

Example: $S = \{(-1, 2, -1)\}$ is linearly independent.

(3) A set of nonzero vectors is **linearly dependent** if and only if **at least one** of the vectors can be expressed as a **linear combination of other vector/s** in the set.

(4) Let $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a set of vectors in $\mathbb{R}^m$ with n elements. If $\mathbf{m < n}$, then the set S is **linearly dependent.**

Example: Let $S = \{(1, 2, -1), (1, 2, 1), (1, -2, -1), (-1, 2, -1)\}$.

Here, $m = 3$ and $n = |S| = 4$.

Here, $m < n$. Hence, S is linearly dependent.

(5) Every **subset** of a linearly independent set is linearly independent.

(6) Every **superset** of a linearly dependent set is linearly dependent.

Procedure to check whether a set is LD or LI

→ Let $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a set of vectors from V.

(1) Form the system $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = 0}$.

(2) Write the corresponding system of linear equations in matrix form as $\mathbf{AX = B}$.

(3) Solve the system and analyze the consistency of the system:

- If the system has **trivial solution**, then S is **LI**.
- If the system has **nontrivial solution**, then S is **LD**.

Remarks

→ To analyze the consistency of the system $AX = B$ **(For Square Matrix only)**

- If $\mathbf{det\,(A) \neq 0}$, then S is **LI**.
- If $\mathbf{det(A) = 0}$, then S is **LD**.

Examples of Method-4: Linear Dependence and Independence

C	1	Is $S = \{(0,0,0),(1,2,3)\}$ is linearly dependent or independent. Why? **Answer: Linearly dependent, it contains zero vector**
C	2	Is $S = \{(2,4,6),(1,2,3)\}$ is linearly dependent or independent. Why? **Answer: Linearly dependent because $\bar{u} = 2\bar{v}$**
C	3	Is $S = \{(1,1),(3,5),(-1,6),(7,2)\}$ is linearly dependent? **Answer: Linearly dependent because $\dim(\mathbb{R}^2) < \lvert S\rvert$**
C	4	Check whether $A = \{(2,1,1),\ (1,2,2),\ (1,1,1)\}$ is subset of $\mathbb{R}^3$ are LD or LI.
C	5	Is $S = \{2 - x + x^2,\ 3 + 6x + 2x^2,\ 2 + 10x - 4x^2\}$ linearly dependent or independent? **Answer: Linearly independent**
C	6	Check $\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, $\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}$ and $\begin{bmatrix}0 & 1\\ 0 & 2\end{bmatrix}$ are linearly dependent or independent in $\mathbb{M}_2$. **Answer: Linearly independent**

Method 5 ⇝ Basis and Dimension

Basis

→ A non-empty subset $S = \{ u_1, u_2, u_3, \dots, u_n \}$ of a vector space V is called a basis of V if:

(1) S is linearly independent.

(2) Span(S) = V.

Procedure to check whether set S is basis of V

→ Let $S = \{u_1, u_2, u_3, \dots, u_n\}$ be a subset of a vector space V.

(1) Write $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = b}$.....(i) – (To check the span)

and $\mathbf{c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_nu_n = 0}$.....(ii) – (To check LI)

Where b is an arbitrary vector of V.

(2) If the linear **system (i)** is **consistent** for all choices of **b** and

the linear **system (ii)** has only **trivial solution**,

then **S** is a **basis** for V.

Remarks

(1) To analyze the consistency of the system AX = B **(For Square Matrix only)**

- If $\mathbf{det\,(A) \neq 0}$, then **S is a basis** of V.
- If $\mathbf{det(A) = 0}$, then **S is not a basis** of V.

(2) A basis is a **minimal spanning set**.

(3) A basis is a **maximal linearly independent set**.

(4) Every vector in a vector space has a unique coordinate representation relative to a basis.

(5) All bases of a vector space contain the same number of vectors.

Dimension of a Vector Space

→ The dimension of a vector space V is defined as the number of vectors in any basis of V. It is denoted by $\mathbf{dim(V)}$.

→ **Examples:**

(1) The vector space $\mathbb{R}^n$ has the standard basis $\{(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1)\}$, which contains n vectors.

Hence, $\dim(\mathbb{R}^n) = n$.

(2) The vector space of all polynomials of degree at most n, $P_n = \{a_0 + a_1x + \cdots + a_nx^n : a_i \in \mathbb{R}\}$, has the basis $\{1, x, x^2, x^3, \ldots, x^n\}$, which contains $n + 1$ vectors.

Therefore, $\dim(P_n) = n + 1$.

(3) The vector space $M_{m\times n}$ of all $m \times n$ matrices have a basis consisting of matrices with exactly one entry equal to 1 and all other entries equal to 0. Since there are $m \times n$ such matrices, $\dim(M_{m\times n}) = mn$.

→ **Remarks:**

(1) If number of elements in given set is less than the dimension of vector space, then given set is never a basis for given vector space.

Example: $S = \{(-1, 2, -1), (1, 1, 8)\} \subset \mathbb{R}^3$. Since there are 2 vectors in S which is less than dimension of $\mathbb{R}^3$. So, S is not a basis for $\mathbb{R}^3$.

(2) If number of vectors in any basis of a vector space V **[dim(V)]** is finite, then V is called finite dimensional. Otherwise, V is called infinite dimensional.

(3) If V is finite-dimensional and U is a subspace of V, then **dim U ≤ dim V**.

(4) If $\dim(V) = n$, then every set containing **more than n** vectors is **linearly dependent**.

(5) If $\dim(V) = n$, then every spanning set with exactly n vectors form a basis of V.

(6) Every finite-dimensional vector space has **infinitely many different bases**.

Examples of Method-5: Basis and Dimension

C	1	Determine whether the following set of vectors forms a basis for $\mathbb{R}^3$ or not. $\{(1, 1, 1), (1, 2, 3), (-2, -1, 1)\}$. **Answer: Yes**
C	2	Show that the set $S = \{A, B, C, D\}$ is a basis for the vector space $\mathbb{M}_{2\times2}$. Where $A = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$, $B = \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}$, $C = \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix}$ and $D = \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}$.
C	3	Show that $S = \{1 - 3x + 2x^2, 1 + x + 4x^2, 1 - 7x\}$ is a basis for $\mathbb{P}_2$.
C	4	Can vectors $u_1 = 3 + x^3, u_2 = 2 - x - x^2, u_3 = x + x^2 - x^3$ and $u_4 = x + x^2$ forms a basis for a vector space $\mathbb{P}_3$? **Answer: No**

C	5	Write a standard basis and find dimension of the following vector spaces. (1). $\mathbb{R}^4$, (2). $\mathbb{P}_5$, and (3). $\mathbb{M}_{2\times3}$.
C	6	Let $\mathbb{M}_2$ be a vector space and S be the subspace spanned by the set of vectors $\left\{\begin{bmatrix}1 & 2\\ 1 & -2\end{bmatrix},\begin{bmatrix}3 & 8\\ 5 & -6\end{bmatrix},\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\right\}$. Find a basis for S and evaluate dim(S). **Answer: Basis** $= \left\{\begin{bmatrix}\mathbf{1} & \mathbf{2}\\ \mathbf{1} & \mathbf{-2}\end{bmatrix},\begin{bmatrix}\mathbf{3} & \mathbf{8}\\ \mathbf{5} & \mathbf{-6}\end{bmatrix}\right\}$, **Dimension** $= \mathbf{2}$

Method 6 ⇝ Row Space, Column Space, and Null Space

Row Space, Column Space and Null Space of a Matrix

→ Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}$ be an $m \times n$ matrix.

→ The vectors $r_1, r_2, \ldots, r_m$ in $\mathbb{R}^n$ are called row vectors of A.

$r_1 = [a_{11} \quad a_{12} \quad \cdots \quad a_{1n}], r_2 = [a_{21} \quad a_{22} \quad \cdots \quad a_{2n}], \ldots, r_m = [a_{m1} \quad a_{m2} \quad \cdots \quad a_{mn}]$

→ The vectors $c_1, c_2, \ldots, c_n$ in $\mathbb{R}^m$ are called column vectors of A.

$$c_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, c_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \ldots, c_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

(1) **Row Space:** The subspace of $\mathbb{R}^n$ spanned by the row vectors of A is called the row space of A.

(2) **Column Space:** The subspace of $\mathbb{R}^m$ spanned by the column vectors of A is called the column space of A.

(3) **Null Space:** The set of all solutions of the homogeneous system of simultaneous linear equations $AX = \mathbf{0}$ is called the null space of the given system.

Procedure to Find the Row Space and its Basis of a Matrix A

→ Let A be an $m \times n$ matrix.

(1) Convert the matrix A into a **row-echelon form** with **leading element '1'**.

(2) Identify all the **non-zero row vectors** in the row-echelon form of A.

(3) Let $B = \{ v_1, v_2, \ldots, v_k \}$ be the **set of all non-zero row vectors** obtained in step (2). Then B forms a basis for the **row space** of A.

(4) The row space **[Row(A)]** of A is the span of the vectors of a basis B.

i.e., $\mathbf{Row(A) = Span\{ v_1, v_2, \ldots, v_k \}}$

→ **Note:** The number of vectors in basis B is called the dimension of the row space of A.
i.e., $\dim(\text{Row(A)}) = \text{rank(A)} = \rho(A) = |\, B \,| = k$.

Procedure to find the Column Space and its Basis for of a Matrix A

→ Let A be an m × n matrix.

(1) Convert the matrix A into a **row-echelon form** with **leading elements '1'**.

(2) Identify the **columns containing leading 1's** in the row-echelon form of A.

(3) Select the **corresponding columns** from the original **matrix A**. Let $B = \{ v_1, v_2, \dots, v_l \}$ be the set of these columns of the original matrix A. Then B forms a basis for the **column space** of A.

(4) The column space of A is the span of the vectors of a basis B.

i.e., $\mathbf{Col(A) = Span\{ v_1, v_2, \dots, v_l \}}$

→ **Note:** The number of vectors in basis B is called the dimension of the column space of A. i.e., $\dim(\text{Col(A)}) = \text{rank of A} = \rho(A) = |\ B\ | = l$.

Procedure to find the Null Space and its Basis

→ Consider the homogeneous system of simultaneous linear equation.

(1) Write the given system in matrix form $\mathbf{AX = 0}$.

(2) Solve the system and obtain the **solution vector X**.

(3) Express X in the form $\mathbf{X = c_1X_1 + c_2X_2 + \cdots + c_pX_p}$, where $c_1, c_2, \dots, c_p$ are arbitrary constants and $X_1, X_2, \dots, X_p$ are fixed vectors.

The set $\mathbf{B = \{ X_1, X_2, \dots, X_p \}}$ forms a **basis for the null space** of the given system.

→ **Note:** The number of vectors in basis B is called the dimension of the null space OR nullity of A. i.e., $\dim(\text{N(A)}) = \text{nullity of A} = \eta(A) = |\ B\ | = p$.

→ **Remarks:**

(1) The null space is the span of the vectors of a basis B, i.e., $\mathbf{N(A) = Span\{X_1, X_2, \dots, X_p\}}$.

(2) For any matrix A, **Rank of A + Nullity of A = number of columns of A**.

(3) If the coefficient matrix A is given directly, then find the null space by solving $\mathbf{AX = 0}$.

(4) Whenever a question asks to find the solution set, solution space, or general solution of a homogeneous system $AX = 0$, it is **equivalent to find** the null space $\mathbf{N(A)}$.

Examples of Method-6: Row Space, Column Space, and Null Space

C	1	Find the basis and dimension for the row and column space for the matrix $A = \begin{bmatrix} 1 & -3 & 4 & -2 & 5 & 4 \\ 2 & -6 & 9 & -1 & 8 & 2 \\ 2 & -6 & 9 & -1 & 9 & 7 \\ -1 & 3 & -4 & 2 & -5 & -4 \end{bmatrix}$. **Answer**: $\{[\mathbf{1\ -3\ \ 4\ -2\ \ 5\ \ 4}], [\mathbf{0\ \ 0\ \ 1\ \ 3\ \ -2\ -6}], [\mathbf{0\ \ 0\ \ 0\ \ 0\ \ 1\ \ 5}]\}, \mathbf{3}$ $\left\{\begin{bmatrix} \mathbf{1} \\ \mathbf{2} \\ \mathbf{2} \\ \mathbf{-1} \end{bmatrix}, \begin{bmatrix} \mathbf{4} \\ \mathbf{9} \\ \mathbf{9} \\ \mathbf{-4} \end{bmatrix}, \begin{bmatrix} \mathbf{5} \\ \mathbf{8} \\ \mathbf{9} \\ \mathbf{-5} \end{bmatrix}\right\}, \mathbf{3}$
C	2	Find a basis for the null space of $A = \begin{bmatrix} 2 & -1 & -2 \\ -4 & 2 & 4 \\ -8 & 4 & 8 \end{bmatrix}$. **Answer**: $\left\{\begin{bmatrix} \mathbf{0.5} \\ \mathbf{1} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix}\right\}$

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